Institut Teknologi Sepuluh Nopember

Selected Topic on Data Science

IDENTIFYING FACTORS THAT DISTINGUISH COMEBACK STUDENTS FROM PERSISTENTLY FAILING STUDENTS THROUGH G1 → G2 → G3 GRADE TRANSITIONS

Mathematics and Portuguese Data, UCI Student Performance Dataset

Group 4

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MOTIVATION / BACKGROUND

1. **Student performance develops** throughout the school year.
2. Final grades alone **do not show students academic progress.**
3. Schools need to identify **which students may recover and when**
4. The UCI Student Performance Dataset contains **grades (G1, G2, G3) and student characteristics.**
5. The main question is: among students **who failed in G1**, **who passed by G3**, and **what characteristics** were associated with recovery?

CHALLENGE

G1 and G2 are **strongly associated** with G3.

Final grades alone **cannot explain when students recover or continue** to struggle.

Analyzing **final grades alone provides limited insight** into students academic progress and recovery.

Schools need to **identify which students may recover from early academic failure and when**.

IDEA

Focus on students **who fail in the first period** (G1 < 10).

Compare students **who recover and pass by G3 (comeback)** with those who continue to fail.

Classify students into four **academic trajectories** based on their G1, G2, and G3 grades.

Identify characteristics associated with academic recovery.

STUDENT ACADEMIC TRAJECTORY

Early Comeback : G1 < 10 → **G2 ≥ 10** → **G3 ≥ 10**

Temporary Comeback : G1 < 10 → **G2 ≥ 10** → G3 < 10

Late Comeback : G1 < 10 → G2 < 10 → **G3 ≥ 10**

Presistent Failure : G1 < 10 → G2 < 10 → G3 < 10

RESEARCH GOAL

1. **Cleaning the Data**
   Check missing values, duplicates, and inappropriate data.
2. **Explore Grade Distributions**
   Understand the patterns of G1, G2, and G3.
3. **Visualize Data Patterns**
   using bar charts, histograms, box plots, scatter plots, and correlation heatmaps.
4. **Identify Key Factors**
   Examine which factors are most strongly associated with final grades.
5. **Analyze Academic Recovery**
   Identify factors that distinguish comeback students from persistent-failure students, then compare the patterns between Mathematics and Portuguese.

DATASET OVERVIEW

Mathematics

school	sex	age	...	G1	G2	G3
GP	F	18	...	5	6	6
...	...	...	...	...	...	...
MS	M	9	...	8	9	9

Portuguese

school	sex	age	...	G1	G2	G3
GP	F	18	...	0	11	11
...	...	...	...	...	...	...
MS	M	18	...	10	11	11

DATASET OVERVIEW

Mathematics

357 analysed

395 initial, 38 excluded

Portuguese

633 analysed

649 initial, 16 excluded

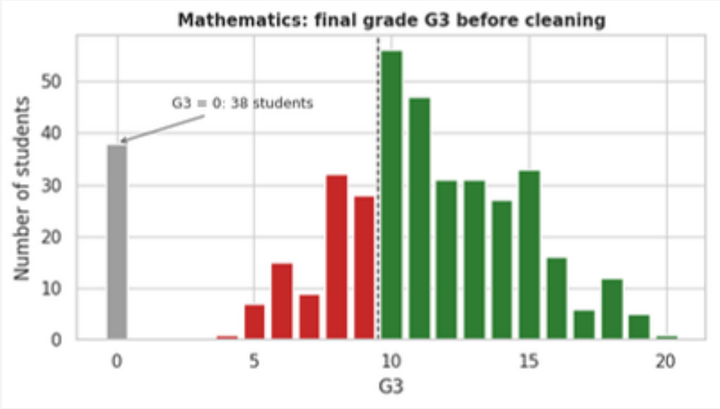
Variable Used : G3 is the final grade. G1 and G2 are previous grades: G1 is used as a control, and G2 as a checkpoint. The 30 background variables are candidate factors.

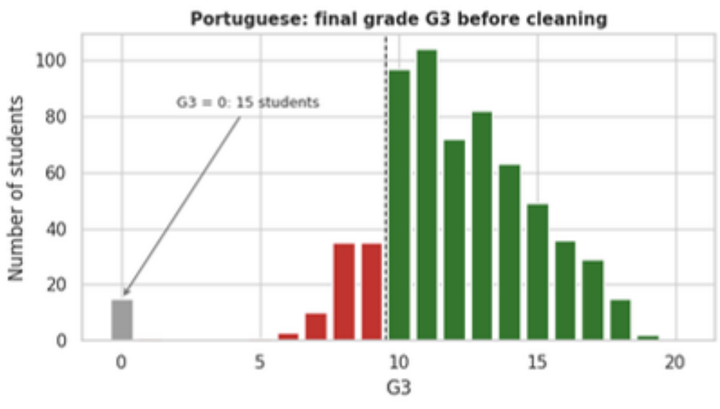
STUDENT BACKGROUND

Data Type	Number of Variable	Description
Binary	13	Two possible values
Nominal	4	Categories without natural order
Ordinal	11	Ordered categories or rating scale
Numeric	2	Quantitative variable
Total	30	

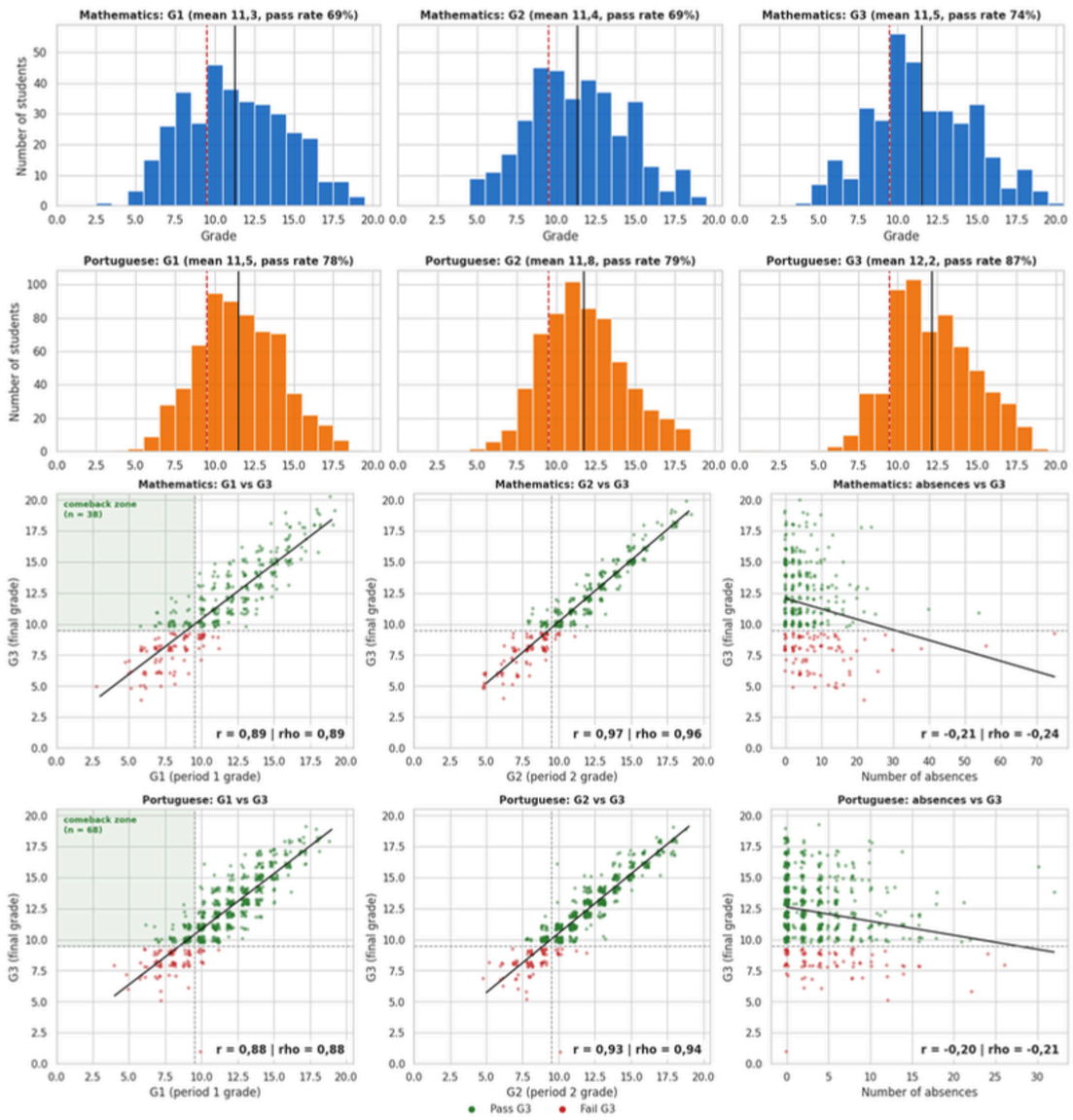
DATA CLEANING

1. **Checked** for missing values, duplicates, and invalid records.
2. **Excluded** records containing a zero in G1, G2, or G3 from the main transition analysis.
3. **Mathematics**: 395 initial → 38 excluded → 357 analyzed.
4. **Portuguese**: 649 initial → 16 excluded → 633 analyzed.
5. Approximately **382 students appear in both subject** datasets.





GRADE DISTRIBUTION AND PREVIOUS GRADES



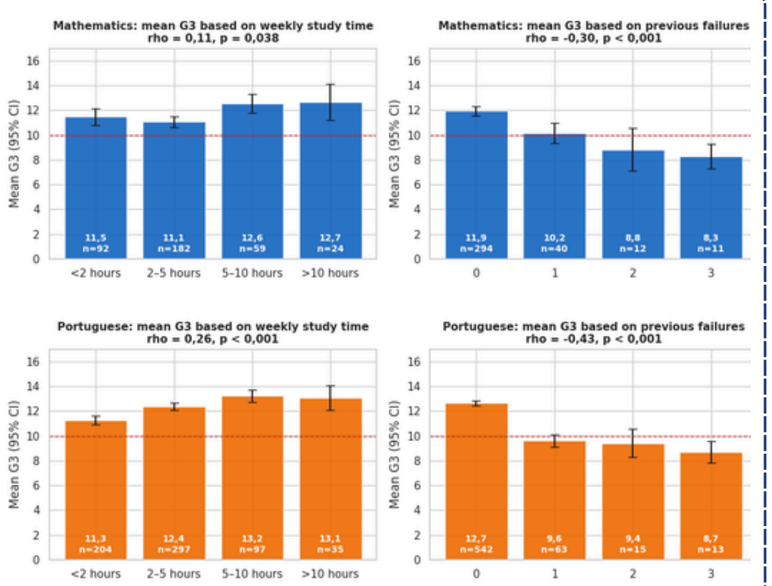
	MATH	PORTUGUESE
Pass rate G1/G3	**69%/74%**	**78%/87%**
Mean G3	**11.5**	**12.2**
r (G1, G3)	**0.89**	**0.88**
r (G2, G3)	**0.97**	**0.93**
Comeback Zone	**38**	**68**

↓

Grades **barely change** from G1 to G3, and **Portuguese has a higher pass rate** (87% vs 74%). **Previous grades are strongly correlated with the final grade**: G1 (r = 0.89 / 0.88) and G2 (r = 0.97 / 0.93), with G2 the closest. Still, **38 Math and 68 Portuguese students recovered from failing G1.**

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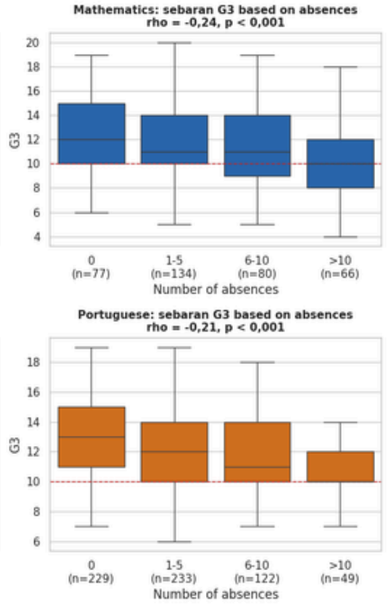
STUDY TIME



BAR CHART

Studying **more than 5 hours a week associateed with only 1–2 points higher**. More study time is linked to slightly higher grades, but does not guarantee a better grade.

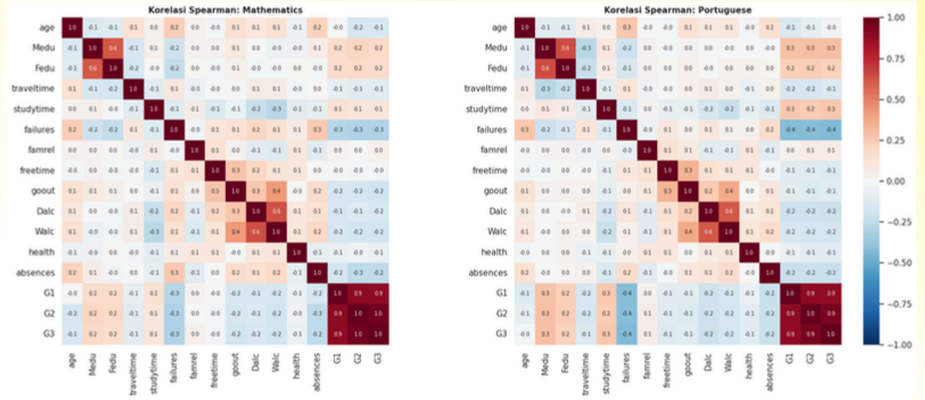
ABSENCES



BOX PLOT

The more often students are absent, the lower their final grade tends to be. The median drops from **12 to 10 (Math)** and **13 to 10 (Port.)**.

CORRELATIONS



CORRELATION HEATMAP

G1, G2 and G3 form a dark red block (about 0.9): early grades almost predict the final grade. **Absences and study time are pale, so each matters much less. Group averages differ, but individual students vary widely, so the link is weak.**

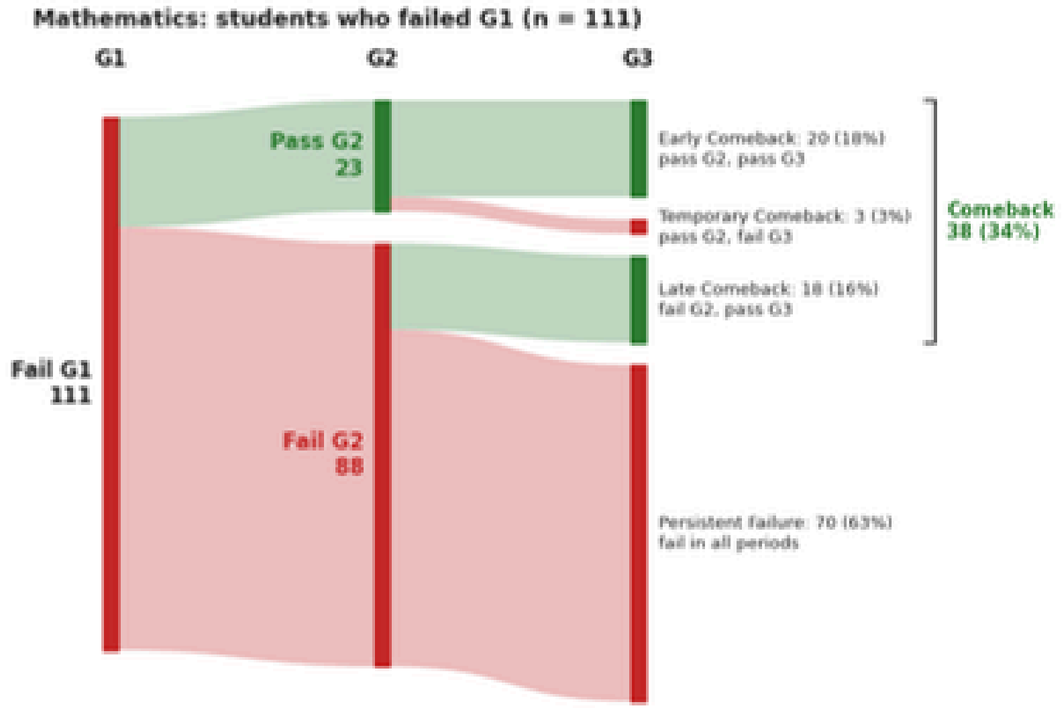
METHOD

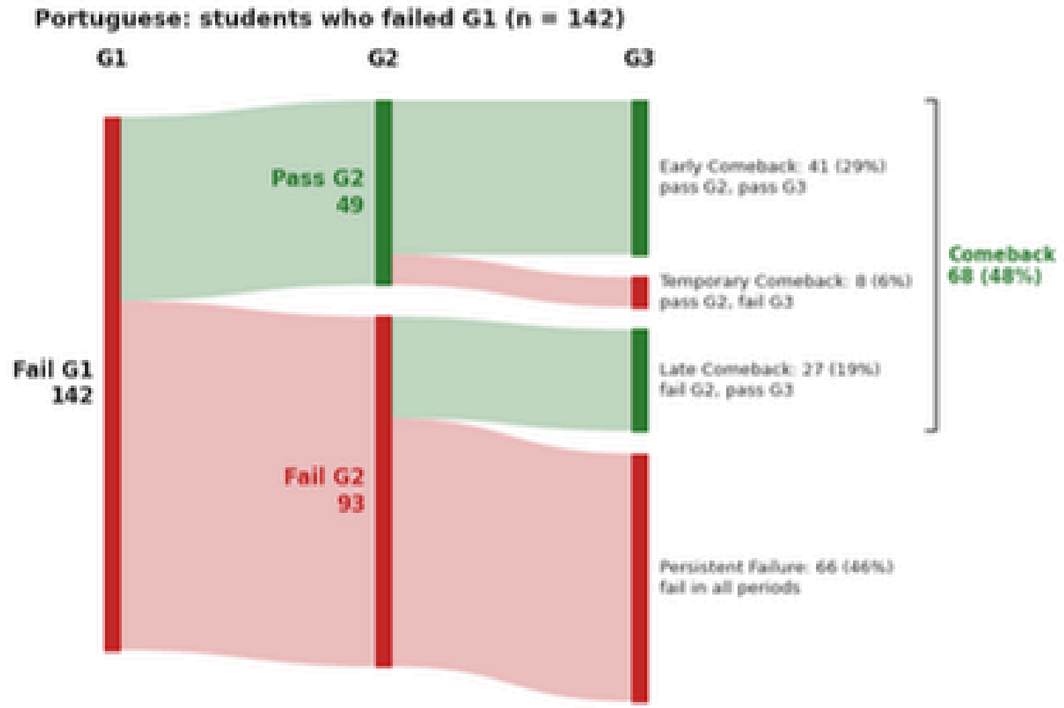
1. Spearman Correlation
Measures how strongly two variables move together (–1 to 1).

2. **Control for G1**
Checks if a factor still matters for students with the same G1.

3. **Multiple-test Correction**
Removes findings that appear by chance (Benjamini–Hochberg).

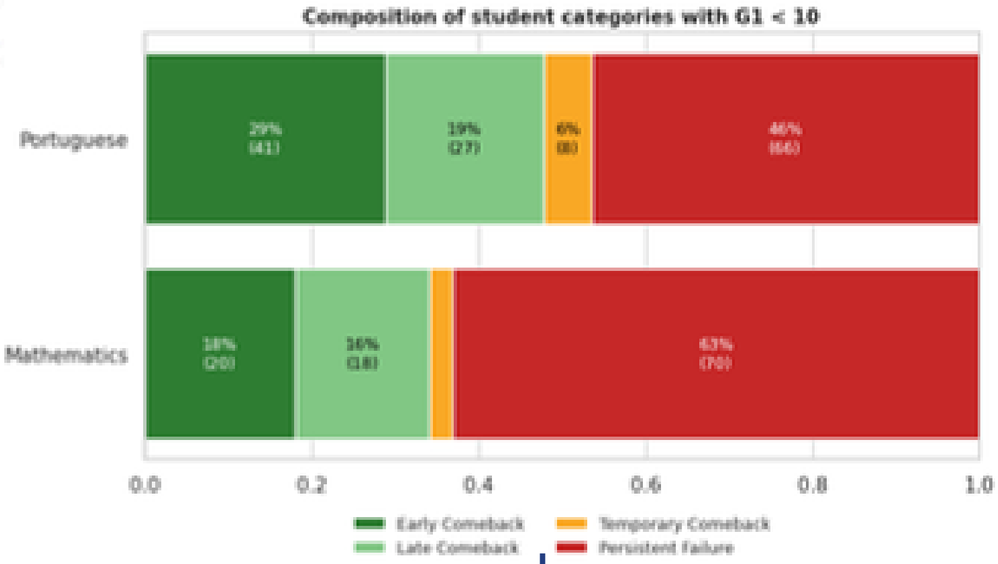
Grade Transition Flow

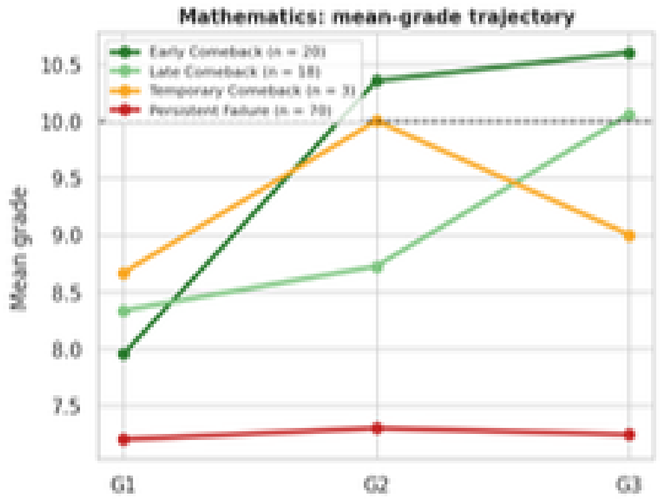


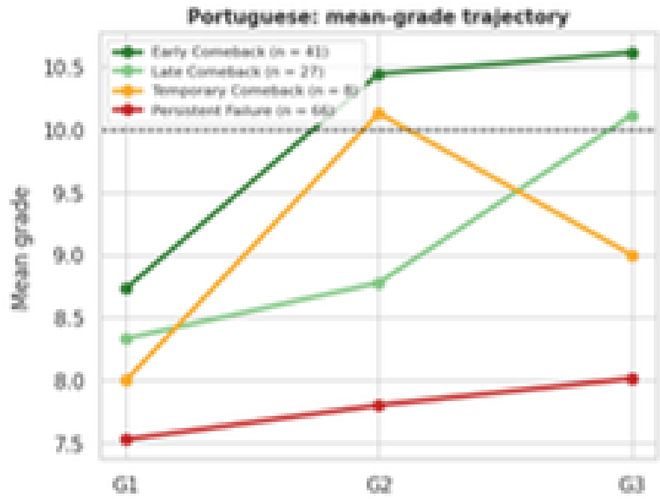


Path (G1 → G2 → G3)	Category	Mathematics (n = 111)	Portuguese (n = 142)
Fail → Pass → Pass	Early Comeback	20 (18%)	41 (29%)
Fail → Fail → Pass	Late Comeback	18 (16%)	27 (19%)
Fail → Pass → Fail	Temporary Comeback	3 (3%)	8 (6%)
Fail → Fail → Fail	Persistent Failure	70 (63%)	66 (46%)

Trajectory Categories: Composition and Mean Grades





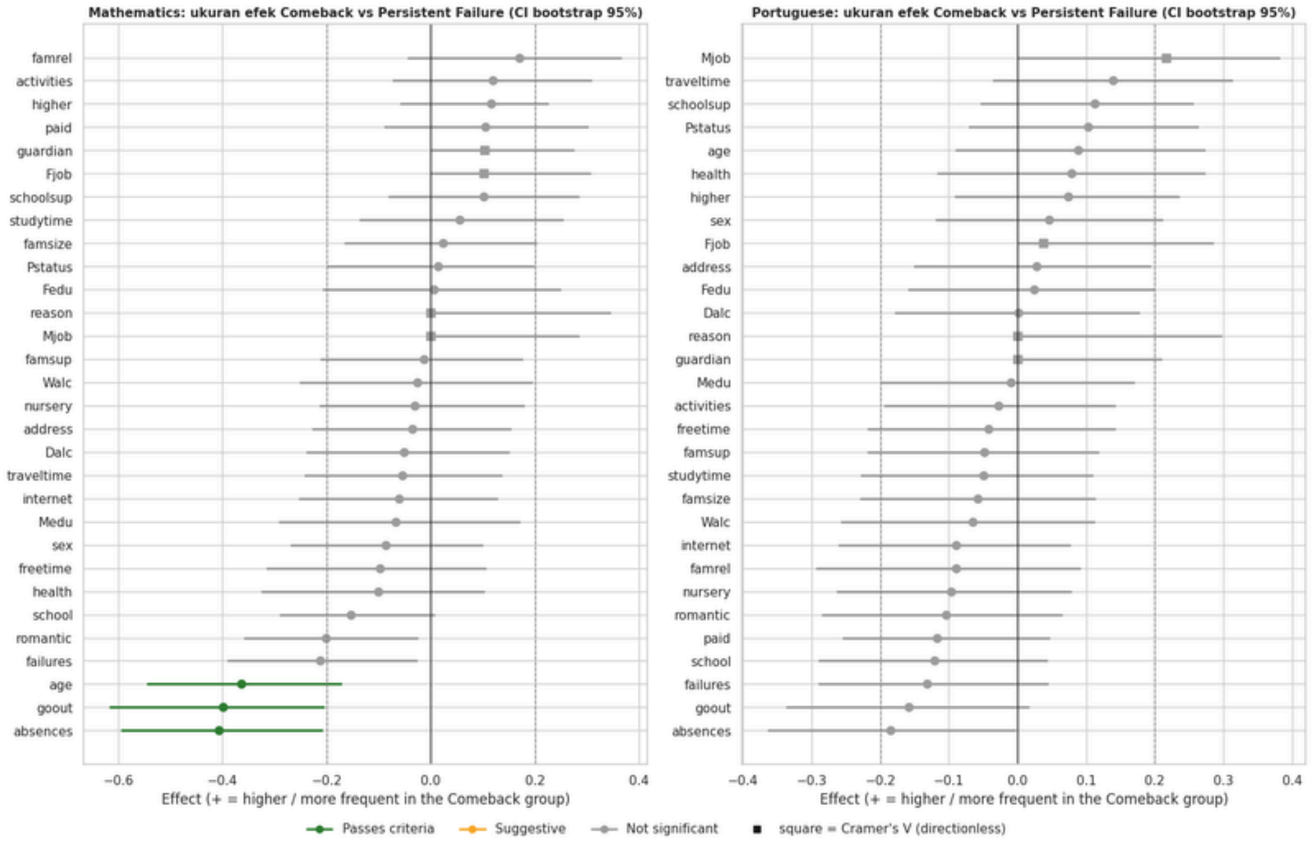


Portuguese has more comebacks; Mathematics has more students who never recover.

Persistent Failure starts lowest (7.2 Math, 7.5 Port.) and stays below 10. Early Comeback rises at G2; Late Comeback rises at G3.

One in three Mathematics students and almost half of Portuguese students recover after failing G1. Those who never recover already start with the lowest grades.

Factors That Distinguish Comeback Students



TOP 3 FACTORS (MATH)

Absences	**4 vs 6.5**
Going out	**2.5 vs 4**
Age	**16 vs 17**

TOP 3 FACTORS (Port.)

Mother's job	**comeback 58% (other) vs 21% (services)**
Absences	**4.3 vs 6.2**
Going out	**3 vs 4**

Method

1. **Test by data type**

Ordinal/numeric: Mann–Whitney U. Binary: Fisher's exact. Nominal: Chi-square. Each gives an **effect size = how big the gap is.**

2. **Correct for 30 tests**

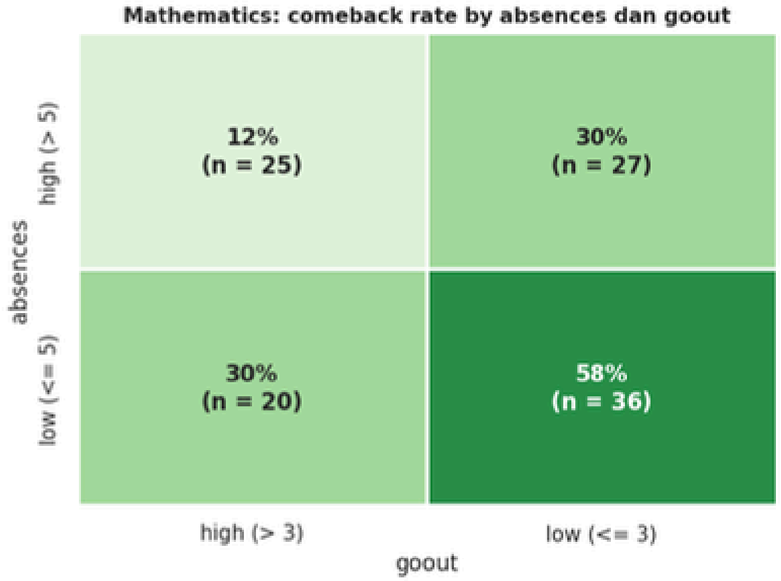
Benjamini–Hochberg correction reduces chance findings. Pass rule: $q < 0.05$ and effect ≥ 0.20.

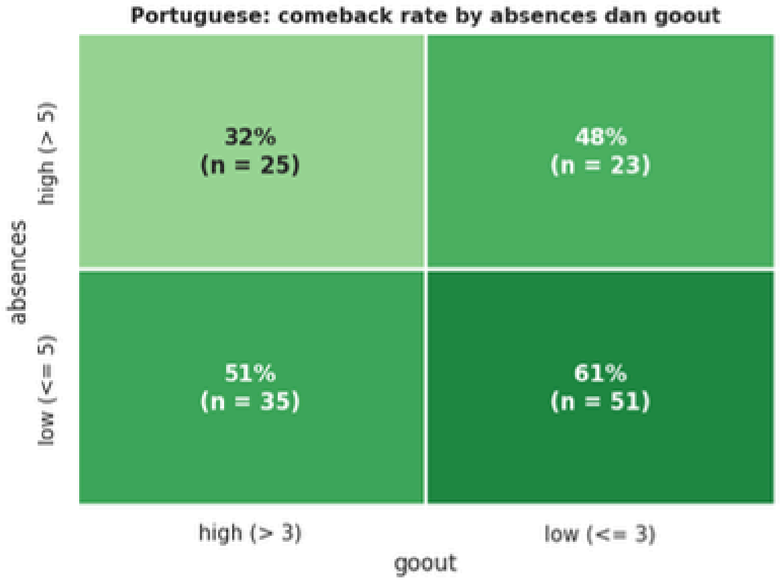
3. **Five robustness checks**

Significant, large enough, holds after controlling for G1, stable in 1,000 bootstrap samples, same direction in the other subject (score 0–5).

SIMPLE RISK PROFILE

Comeback rates among students who failed G1, grouped by absences and going-out frequency





Overall Finding:
Lower absences and going-out frequency **were associated with higher comeback rates** in both subjects.

Comeback rate **Mathematics**
58% vs **12%**

Comeback rate **Portuguese:**
61% vs **32%**

CONCLUSION

G1 and G2 strongly predict the final grade ($r \approx 0.9$). Study time, absences and past failures matter much less.

Study time, absences and past failures matter much less. After controlling for G1, only age and absences (Math) and age and failures (Port.) remain.

34% of Math and 48% of Portuguese students who failed G1 still passed G3.

In Math, they are absent less, go out less and are younger. In Portuguese, the same signs appear but are too weak to confirm.

RECOMMENDATION

Monitor absences and going out after G1, and give extra support to students still failing at G2.



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THANK YOU



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